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SMART HOME SYSTEM

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Abstract

- Smart Home system using STM32 and ESP32 for embedded control and IoT connectivity.
- STM32 runs RTOS to manage sensors, actuators, and real-time control tasks.
- ESP32 handles Wi-Fi communication and MQTT data exchange with the cloud server.
- A robust backend handles data storage and processing, enabling seamless communication between hardware and user interfaces.
- Web dashboard provides monitoring, remote control, and automatic operation features.

Motivation and Objectives

- Motivation
  - Existing Smart Home solutions are costly and mainly designed for high-end houses, limiting use in apartments and small homes.
  - Fire risks in dense urban areas require low-cost systems with real-time detection and cellular-based alerting.
- Objectives
  - Design a low-cost Smart Home system suitable for apartments and small residential buildings.
  - Achieve real-time and responsive system operation using FreeRTOS to ensure reliable control, monitoring, and user experience.

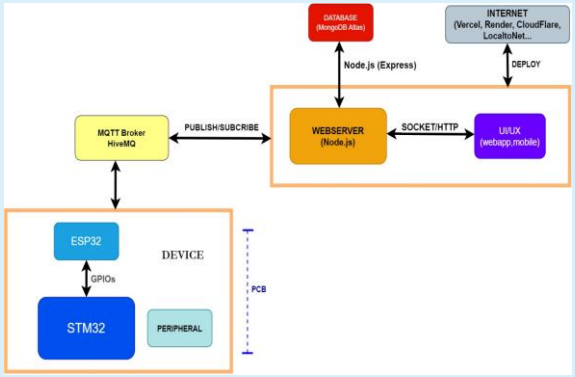
Software Methodology

Transfer Data between Hardware and Web

Database for User, Data storage

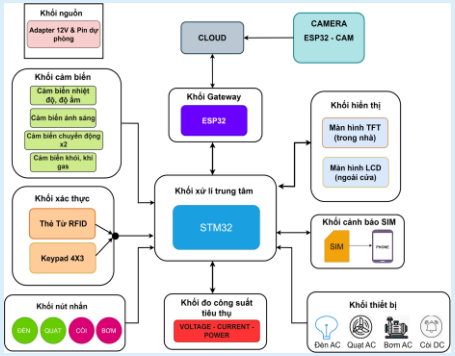
Language for backend and web interface.

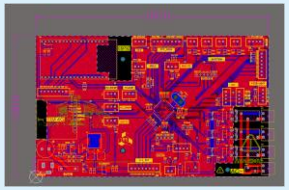
- The ESP32 exchanges data with STM32 and publishes system information to the MQTT broker.
- The MQTT broker forwards messages to a Node.js and Express server for processing.
- The backend stores system states and events in MongoDB Atlas.
- The web-based user interface communicates with the backend via HTTP and WebSocket for real-time monitoring and control.



Hardware Methodology

- The system is implemented on a custom-designed PCB developed using Altium Designer.
- The PCB integrates STM32, ESP32, power management, sensors, and communication modules.
- Hardware design focuses on stability, reliability, and real-time operation.





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References

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